## Name: Surya Sutra

***a Solar Microgrid Optimizer for Equitable Energy Access***

## Team name: Algoholics

## Concept Note:

### Problem Statement

Delhi-NCR faces a dual energy crisis: unreliable electricity access and extreme air pollution caused by diesel generators. With over 4,600 outages annually and 19–21% grid losses, the energy inequity is glaring. Low-income households endure frequent blackouts, relying on polluting diesel backups while affluent zones receive uninterrupted supply. Current microgrid solutions focus only on solar efficiency, neglecting equitable access. This results in solar infrastructure concentrating in high-income zones, perpetuating injustice.

### Target Users

| User Type | Role | Geography | Needs |
|---|---|---|---|
| Municipal Energy Planners | Design 5-year solar roadmaps | Delhi, NCR cities | Fast, accurate site selection |
| DISCOM Grid Managers | Peak load balancing, energy dispatch | Tata Power, BSES, Adani (8.9M households) | Real-time optimization |
| Solar Cooperatives & NGOs | Expand community energy access | SELCO, Greenkraft, housing societies | Equity-first site selection transparency |
| Low-Income Communities | Ensure transparent site selection | 50M+ urban poor in Delhi’s informal settlements | Direct benefit verification |
| Developers & Commercial | Monetize rooftops and meet ESG goals | Residential, commercial sectors | ROI insights and carbon credit estimation |

### Importance

- **Climate Impact**:India’s 2030 Nationally Determined Contribution (NDC) targets 175 GW of solar energy—achieving this goal equitably requires data-driven planning.
- **Social Justice**: Nearly 40% of Delhi’s population lives in slums. Energy access must be a universal right, not just a privilege for the wealthy.
- **Economic Resilience**: Distributed solar microgrids can reduce peak load by 15–30%, cutting grid management costs by ₹500 crores/year and helping prevent power outages.

### Why Data, AI, and Digital Technology Matter

1. **Data Advantage:** Manual rooftop assessments cost around **₹50,000 per building**,

while satellite imagery and GIS analysis reduce the cost to **₹500 per building**—making it **100× faster and 98% cheaper**.

2. **AI + Optimization Logic:** Traditional solar rollouts favor "sunniest rooftops," ignoring vulnerable communities. Our **multi-objective optimization algorithm** incorporates both **solar potential** and **equity weightage**.
3. **Digital Dashboard:** The interactive tool enables **real-time tradeoff simulations**. For example, setting **equity weight to 70%** increases **equity score to 68%**, while coverage might drop to 71%—this transparency empowers data-backed policymaking.

## Prototype Design & Workflow:

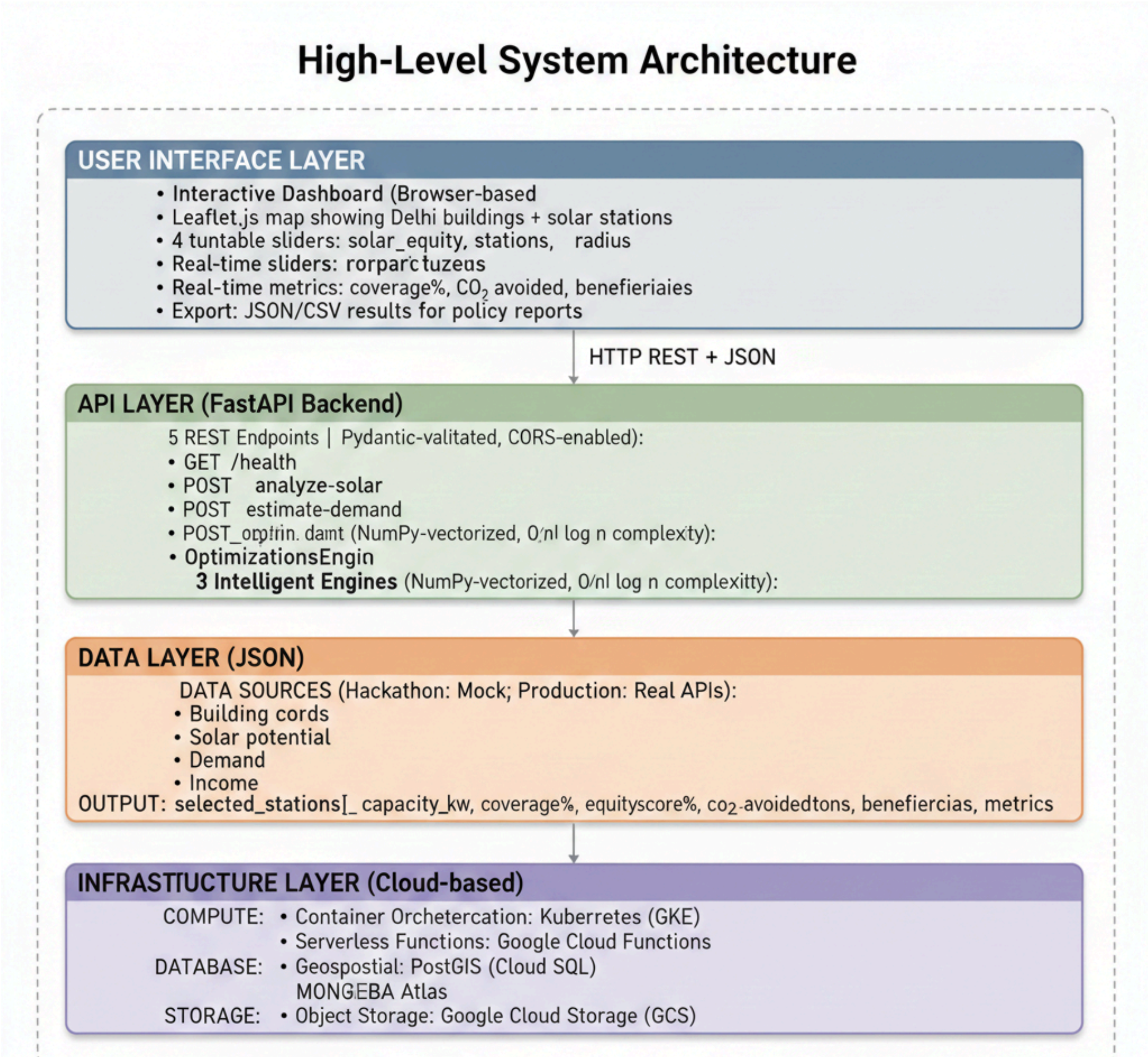


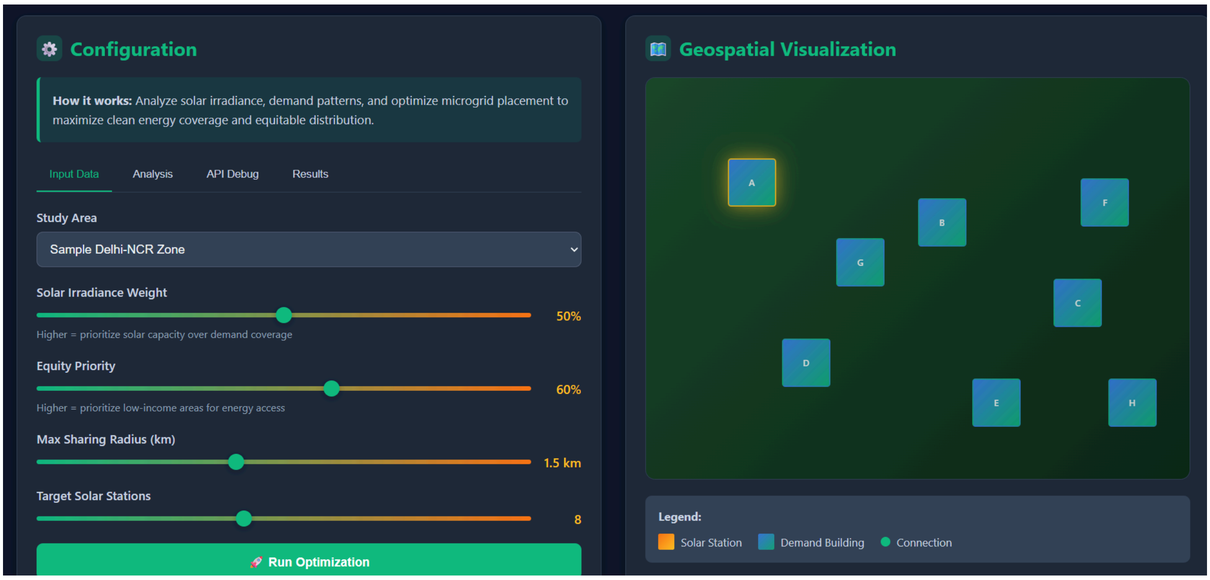

Figure 1: 4-Layer System Stack

### Key Algorithms

- **Solar Score** = (irradiance x 0.5) + (shading x 0.3) + (efficiency x 0.2)
- **Demand Score** = building type × income factor × peak multiplier
- **Composite Score** = (solar_score x solar_weight) + (equity_score x equity_weight)
- **Low-income buildings get higher equity priority** (default 0.9)

*for the hackathon prototype, we use 8 representative buildings with realistic parameters. The algorithm is designed to scale to 500+ buildings with real data.*

### User Journey




**Planner opens the dashboard** – They are presented with a map showing the Delhi-NCR study area.

**Configuration adjustment** – User sets preferences, like 50% solar priority, 60% equity priority, and 8 target solar stations.

**Runs optimization** – With a click, the backend calculates weighted scores for all buildings based on solar potential and equity factors.

**Views results** – The selected solar station buildings are highlighted on the map, along with detailed coverage and equity impact metrics.

**Explores trade-offs** – By adjusting priority sliders, planners can re-run the model and immediately observe how the station selection and impact metrics shift.

**Exports for deployment** – Finalized outputs, including station coordinates and impact data, can be downloaded as JSON for use in planning or integration.

## Early Prototype Model

***Dashboard Visualization*** ***(static version for demo, backend not connected)***

our working prototype enables planners to simulate real-world solar microgrid deployment scenarios across Delhi-NCR. The interface includes:

### Interactive Configuration Panel

- **Study Area Selector** – Choose focus zones within Delhi-NCR.
- **Solar Irradiance Weight Slider** – Adjust importance of solar-rich rooftops (0–100%).
- **Equity Priority Slider** – Emphasize underserved, low-income areas (0–100%).

- **Sharing Radius Control** – Set max energy sharing distance (0.5–3 km).
- **Target Stations Selector** – Specify number of solar hubs (3–15).

### Geospatial Map

- Buildings displayed as clickable polygons.
- Selected **solar station buildings** are highlighted using color gradients.
- **Connection lines** show power distribution to demand buildings.
- Clicking a building reveals attributes: type, solar potential, income level.

### Results Metrics

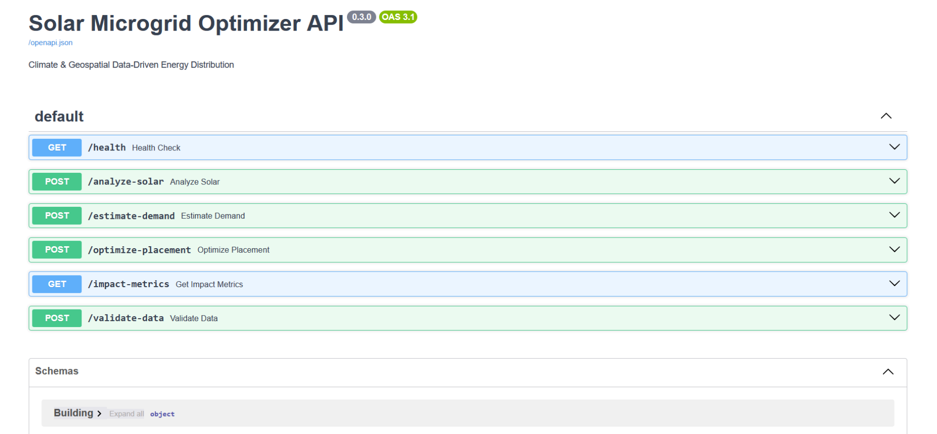
- **Solar Stations Selected** – IDs of optimized source locations.
- **Total Capacity** – Combined power generation in kW.
- **Coverage** – % of total regional energy demand met.
- **Equity Score** – % of low-income zones served.

### Impact Simulation

| **Metric** | **Value** |
|---|---|
| $CO_2$ Avoided (per year) | 10.3 tons |
| Diesel Replaced | 1 Generator |
| Peak Load Reduction | 87% |
| Low-Income Beneficiaries | 4,600 people |
| Savings on Diesel | ₹2.5 L/year |

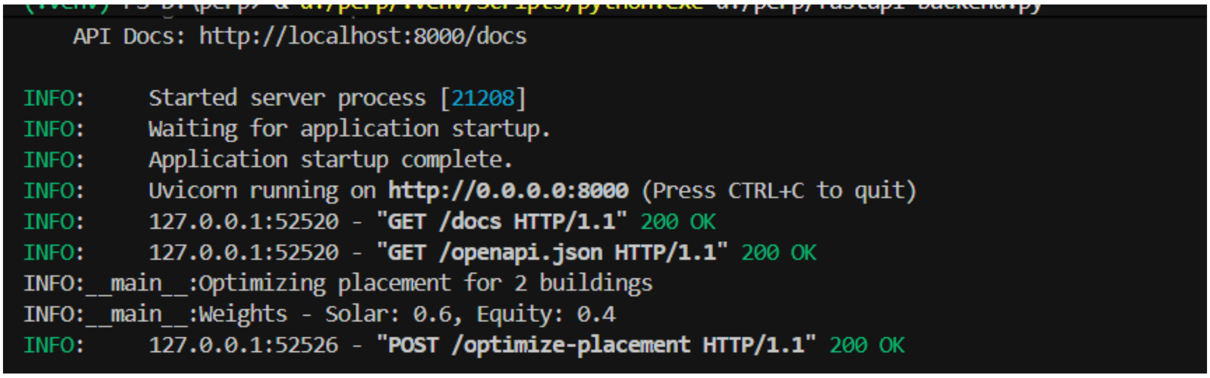
## 3.2 Backend API (Functional)

The FastAPI-powered backend is fully operational and exposed via Swagger UI:




| **Endpoint** | **Purpose** |
|---|---|

| | |
|---|---|
| `GET /health` | Ping endpoint for connection test |
| `POST /analyze-solar` | Calculates solar suitability scores |
| `POST /estimate-demand` | Computes per-building energy demand |
| `POST /optimize-placement` | Runs optimization for solar station placement |
| `POST /validate-data` | Pre-checks uploaded data for integrity |

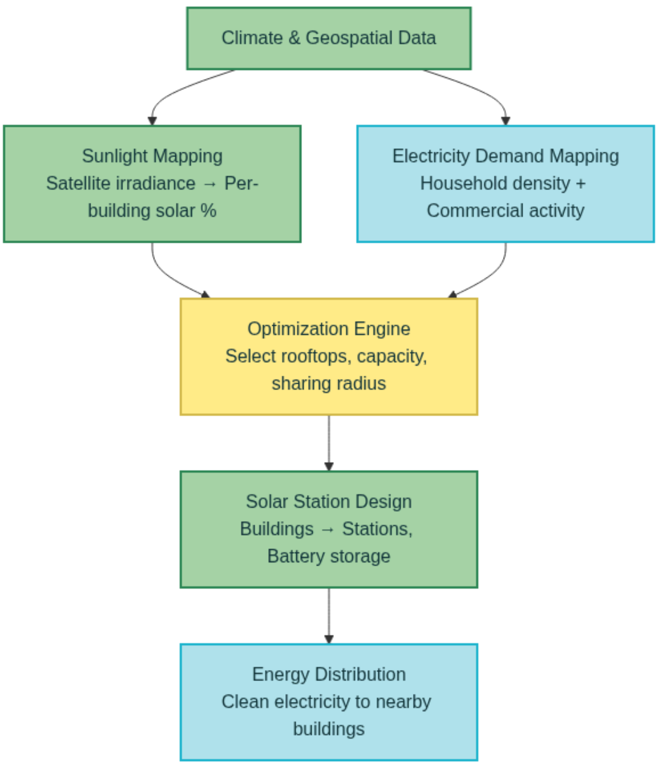


```
    API Docs: http://localhost:8000/docs

INFO:     Started server process [21208]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     127.0.0.1:52520 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:52520 - "GET /openapi.json HTTP/1.1" 200 OK
INFO:__main__:Optimizing placement for 2 buildings
INFO:__main__:Weights - Solar: 0.6, Equity: 0.4
INFO:     127.0.0.1:52526 - "POST /optimize-placement HTTP/1.1" 200 OK
```

*Backend working.*

- **Interactive Dashboard:** `Dashboard.html`
- **Live API Documentation:** Accessible at `http://localhost:8000/docs`
- **System Architecture:**



## Business & Sustainability Snapshot

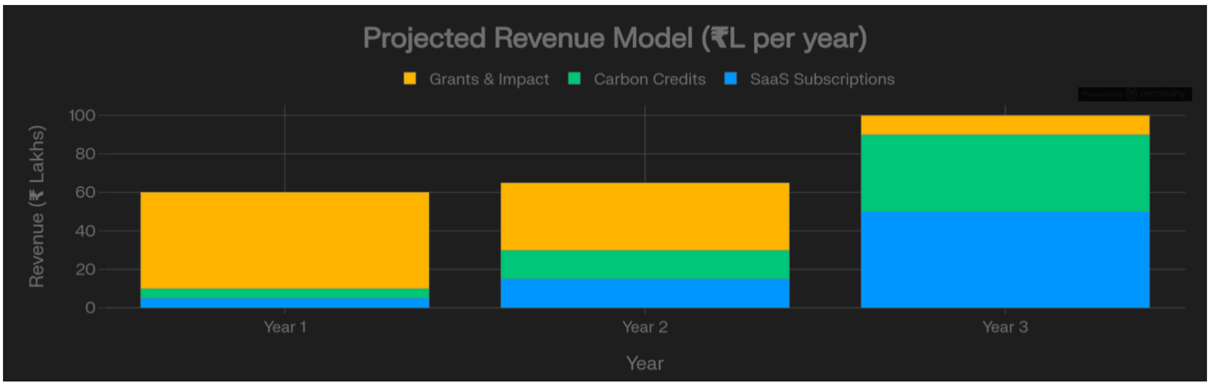
### Customer Segments

| User Segment | Description | Willingness to Pay |
|---|---|---|
| DISCOMs | Delhi electricity distributors like BSES, Tata Power DDL | High – Reduces T&D losses, boosts grid reliability |
| Municipal Corporations | Local bodies (NDMC, SDMC, DDA) managing public energy planning | Medium – Budget-constrained but mission-aligned |
| CSR Funders | Corporates fulfilling ESG mandates under CSR obligations | High – Strong alignment with SDG/ESG metrics |
| Community Cooperatives | RWAs and local societies in underserved zones | Low – Require government or NGO subsidies |

### Value Proposition

| **Legacy Planning** | **Surya Sutra** |
|---|---|
| Manual rooftop surveys (₹50K/site) | Automated geospatial and equity-informed analytics |
| Focuses only on solar potential | Balances solar suitability *and* community need |
| Weeks of static feasibility reporting | Real-time dashboard-based simulation |
| No traceability or transparency | Every decision backed by visual + numeric metrics |
| Ignores distribution equity | 40% energy auto-reserved for low-income users |

### Revenue Model



**Year 1: Pilot Phase**

- **Pricing**: Free deployment in partnership with 3–5 Delhi-NCR DISCOMs or municipalities
- **Revenue Model**: Grants (e.g., CCF-GEF, USAID, Bill & Melinda Gates Foundation)
- **Objective**: Validate outcomes, quantify impact ($CO_2$, equity)

**Year 2–3: Scaling Phase**

- **Pricing**: ₹5–10 lakh/year per utility (SaaS license)
- **Model**: Subscription-based access for utilities, government bodies
- **Revenue**: Annual recurring licensing fees

**Year 3–5: Platform Phase**

- **Pricing**: ₹500–₹2000 per site for API/analysis use
- **Model**: Open platform for solar developers, CSR, cooperatives
- **Revenue**: Transactional fees, data analytics, premium features

**Partnerships**

| Type | Examples | Role in Ecosystem |
|---|---|---|
| **DISCOMs** | BSES Yamuna, Tata Power DDL | Pilot testing, operational data, deployment |
| **Government** | MNRE, Delhi Energy Dept | Funding, policy integration, long-term scaling |
| **NGOs** | CEEW, Prayas Energy Group | Ground validation, equity scoring, outreach |
| **Tech Backers** | Google.org, Microsoft AI for Earth | Cloud infra, GIS/AI support |
| **Academic** | IIT Delhi, IIIT-H | Algorithm validation, impact studies |

## Impact Snapshot

**Pilot (8 Stations, 45kW total)**

- 4,600 low-income beneficiaries
- 10.3 tons $CO_2$ avoided/year
- 87% peak load reduction
- 1 diesel generator replaced
- ₹2.5L/year savings in avoided diesel

**Scaled Projection (100 Buildings)**

- 115 tons $CO_2$/year avoided
- 50,000+ beneficiaries

- 4x capacity, breakeven in 3.8 years

**SDG Alignment**

- SDG 7: Affordable Clean Energy
- SDG 10: Reduced Inequalities
- SDG 11: Sustainable Cities
- SDG 11: Sustainable Cities
- SDG 13: Climate Action

**Key Assumptions**

- **Solar Efficiency**: 18% (typical mono-PERC panels)
- **$CO_2$ Intensity**: 0.62 kg/kWh (India national grid average)
- **Residential Load**: 1.5 kWh/day/household
- **Battery Backup**: 4–6 hours standard for low-cost setups
- **Transmission Losses**: 8% average grid loss factored in
- **Equity-First Design**: 40% energy reserved for low-income; algorithm enforces this by design

GitHub Repo : [https://github.com/nush-13/Climate-Data-Hackathon](https://github.com/nush-13/Climate-Data-Hackathon)